

EXAM 02582 2025 Computational data analysis

02582 Computational Data Analysis

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Technical University of Denmark

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Course Title: Computational Data Analysis

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Aids allowed: All written aids allowed – no internet

Exam Duration: 4 hours

Weighting: 60%

1. In the bias-variance decomposition of prediction error, which component is not directly affected by model complexity?
 - (a) Variance
 - (b) Bias
 - (c) Irreducible error
 - (d) Training error
 - (e) Expected prediction error
2. The kernel trick in SVM allows:
 - (a) Automatic feature selection
 - (b) Computation in high-dimensional feature space without explicit transformation
 - (c) Better regularization
 - (d) Classification without using margin-defining data samples
 - (e) Faster training
3. Which of the following methods is not matrix factorization-based?
 - (a) NMF
 - (b) PCA
 - (c) ICA
 - (d) K-means
 - (e) Archetypal Analysis
4. Which of the following is an implicit assumption behind the validity of standard cross-validation estimate?
 - (a) The training and test sets must be of equal size.
 - (b) The response variable must be normally distributed.
 - (c) The data are independently and identically distributed (IID).

- (d) The model must be linear in parameters.
 - (e) The folds must have the same mean response value.
5. Which of the following best distinguishes information criterion-based model selection (e.g., AIC/BIC) from cross-validation, with respect to their assumptions?
- (a) AIC/BIC assumes a correctly specified likelihood model, while cross-validation makes fewer assumptions about the data-generating process.
 - (b) Cross-validation assumes the model is linear and errors are normally distributed, while AIC/BIC does not.
 - (c) AIC/BIC provides unbiased estimates of out-of-sample error, while cross-validation estimates training error.
 - (d) Cross-validation requires large samples to be valid, while AIC/BIC can be used on any data size without issue.
 - (e) AIC/BIC rely on data splitting like cross-validation, but are computationally more expensive.
6. When is nested cross-validation preferred over standard (non-nested) cross-validation?
- (a) When you have a very large dataset and want to reduce computation time.
 - (b) When the model has no tunable parameters.
 - (c) When you want an unbiased estimate of generalization error after hyperparameter tuning.
 - (d) When you're testing whether the response variable is normally distributed.
 - (e) When cross-validation folds are randomly sampled with replacement.
7. Which of the following best explains why Ridge regression tends not to perform feature selection?
- (a) It uses L1 regularization
 - (b) It removes collinear features before fitting the model
 - (c) It shrinks coefficients but doesn't set them to zero
 - (d) It standardizes features before modelling
 - (e) It excludes features with low variance by default.
8. Which model selection criterion penalizes model complexity the most as $N \rightarrow \infty$? N is the sample size.
- (a) AIC
 - (b) BIC
 - (c) Cp
 - (d) Cross-validation
 - (e) Adjusted R^2
9. In LDA, the decision boundary is linear because:
- (a) It uses polynomial basis functions
 - (b) It assumes equal class priors
 - (c) It minimizes squared loss
 - (d) It fits a linear regression
 - (e) It assumes equal class covariances

10. Which of the following statements best distinguishes the Benjamini–Hochberg (BH) procedure from the Bonferroni correction in multiple hypothesis testing?
- (a) BH controls the expected proportion of false discoveries, while Bonferroni controls the probability of at least one false discovery.
 - (b) BH controls the family-wise error rate, while Bonferroni controls the false discovery rate.
 - (c) BH adjusts p-values to be more conservative than Bonferroni.
 - (d) BH is only valid when all null hypotheses are true, while Bonferroni is valid regardless.
 - (e) BH requires all p-values to be normally distributed.
11. Which of the following best explains why ICA can recover sources that PCA cannot?
- (a) ICA assumes orthogonal components, while PCA assumes independence.
 - (b) ICA identifies components with maximal variance, while PCA minimizes kurtosis.
 - (c) ICA maximizes non-Gaussianity to find statistically independent components, while PCA finds uncorrelated directions.
 - (d) ICA removes noise using eigenvalue shrinkage, which PCA cannot do.
 - (e) ICA rotates the principal components to align with the class labels.
12. Consider a fully connected feedforward neural network with the following architecture- Input layer: 3 nodes; Hidden layer 1: 4 nodes (ReLU activation); Hidden layer 2: 2 nodes (ReLU activation); Output layer: 1 node (Linear activation). **Assume** that each layer includes a bias term. You are given a single input vector $X \in \mathbb{R}^3$. How many scalar parameters (weights and biases) does the network have in total?
- (a) 21
 - (b) 25
 - (c) 27
 - (d) 29
 - (e) 31
13. Which of the following statements most correctly highlights a key difference between K-means clustering and Gaussian Mixture Models (GMMs)?
- (a) K-means allows clusters to have different covariance structures, while GMM assumes identical spherical clusters.
 - (b) K-means assigns soft cluster memberships, while GMM assigns hard labels only.
 - (c) GMM models the data distribution probabilistically, allowing for elliptical clusters, while K-means minimizes squared Euclidean distance to centroids.
 - (d) GMM is more robust to outliers than K-means because it uses medians instead of means.
 - (e) K-means uses likelihood maximization for parameter estimation, while GMM does not.
14. Given a centered data matrix $X \in \mathbb{R}^{100 \times 2}$, and its empirical covariance matrix has eigenvalues $\lambda_1 = 6$, $\lambda_2 = 2$, what fraction of total variance is explained by the first principal component?
- (a) 25%
 - (b) 50%
 - (c) 60%
 - (d) 75%

- (e) 80%
15. Which of the following correctly characterizes a fundamental difference between K-medoids and K-means clustering?
- (a) K-medoids minimizes the sum of squared distances, while K-means minimizes absolute distances.
 - (b) K-medoids selects centroids that may lie outside the data cloud, while K-means does not.
 - (c) K-medoids is less robust to outliers than K-means.
 - (d) K-medoids is more robust to outliers than K-means.
 - (e) K-medoids assumes normally distributed features, unlike K-means.
16. Which of the following is a defining characteristic of Non-negative Matrix Factorization (NMF) when applied to a data matrix $X \in \mathbb{R}^{n \times p}$?
- (a) NMF guarantees a unique factorization for all datasets.
 - (b) NMF seeks matrices W and H such that $X \approx WH$, with all entries of W and H constrained to be non-negative.
 - (c) NMF seeks matrices W and H such that $X \approx WH$, with orthogonality constraints on W .
 - (d) NMF seeks matrices W and H such that $X \approx WH$, with the only assumption that all entries of X to be non-negative.
 - (e) NMF requires the input data matrix X to be square and symmetric.
17. Which of the following best describes a defining characteristic of Archetypal Analysis (AA) when applied to a dataset $X \in \mathbb{R}^{n \times p}$?
- (a) Archetypes are orthogonal directions that capture the maximum variance in the data, and each data point is approximated by a weighted mixture of these archetypes.
 - (b) Archetypes are cluster centroids that minimize squared distances to data points.
 - (c) Archetypes are constructed as weighted combinations of extreme observations, and each data point is approximated by a weighted mixture of these archetypes.
 - (d) Archetypal Analysis assumes features follow a multivariate Gaussian distribution.
 - (e) Archetypal Analysis decomposes the data using sparse binary encodings.
18. Why is it important to apply feature standardization *within* each fold of a cross-validation procedure, rather than using statistics from the entire dataset?
- (a) Standardizing within each fold reduces the variance of cross-validation error estimates.
 - (b) Using full-data statistics can cause information leakage from the test fold into the training process.
 - (c) Standardization is unnecessary if the model is linear.
 - (d) Performing standardization within each cross-validation fold speeds up model training.
 - (e) Standardizing within folds ensures features are normally distributed in each fold.
19. You are shown a plot of Mean Squared Error (MSE) against model complexity. Using the one-standard error rule, which model should be selected?
- (a) Model with complexity = 5
 - (b) Model with complexity = 6
 - (c) Model with complexity = 4

- (d) Model with complexity = 1
 - (e) Model with complexity = 10
20. What is the optimal number of clusters, from the silhouette plot?
- (a) 3
 - (b) 4
 - (c) 5
 - (d) 6
 - (e) 7
21. Discuss LDA and GMM in terms of the following a) Explain and contrast the probabilistic assumptions between them. b) Describe and contrast how model fitting is performed. c) Highlight key differences in their goals, supervision, and use of labels, etc. d) Discuss how each model handles class overlap and latent structure.
22. You are given a dataset consisting of time series biosignals measured by a wearable device, including blood volume pressure (BVP), skin temperature, and heart rate (HR). Data has been collected from 16 individuals under three conditions (rest, running, social media) at four different time points throughout the year (representing seasonal variation). This results in 192 observations ($16 \text{ individuals} \times 3 \text{ activities} \times 4 \text{ seasons}$). Explain and justify how you would design your datasets (training, validation, test) to estimate the expected prediction error if a) Our goal is to develop a personalized model to predict the stress of a specific individual (eg: your smart watches providing vital insights on your future activity level)? b) Our goal is to develop a generalized model to predict the stress of a new individual. Also, discuss the trade-offs between the above two model types and comment on which model type is more appropriate to deploy in the clinic to aid mental health experts with new patients.